

SOLIDWORKS API (.NET VBA C#) — Reference (189/337): IModelDoc2 (part 12 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 12 of 17) (Type)

SOLIDWORKS API Help

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InsertSheetMetalHem Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalHem Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalHem](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalHem( _  
    ByVal Type As System.Integer, _  
    ByVal Position As System.Integer, _  
    ByVal Reverse As System.Boolean, _  
    ByVal DLength As System.Double, _  
    ByVal DGap As System.Double, _  
    ByVal DAngle As System.Double, _  
    ByVal DRad As System.Double, _  
    ByVal DMiterGap As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Type As System.Integer  
Dim Position As System.Integer  
Dim Reverse As System.Boolean  
Dim DLength As System.Double  
Dim DGap As System.Double  
Dim DAngle As System.Double  
Dim DRad As System.Double  
Dim DMiterGap As System.Double
```

```
instance.InsertSheetMetalHem(Type, Position, Reverse, DLength, DGap, DAngle, DRad, DMiterGap)
```

C#

```
void InsertSheetMetalHem(  
    System.int Type,  
    System.int Position,  
    System.bool Reverse,  
    System.double DLength,  
    System.double DGap,  
    System.double DAngle,  
    System.double DRad,  
    System.double DMiterGap  
)
```

C++/CLI

```
void InsertSheetMetalHem(  
& System.int Type,  
& System.int Position,  
& System.bool Reverse,  
& System.double DLength,  
& System.double DGap,  
& System.double DAngle,  
& System.double DRad,  
& System.double DMiterGap  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Position
Reverse
DLength
DGap
DAngle
DRad
DMiterGap

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalHem](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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InsertSheetMetalJog Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalJog Method (IModelDoc2)

Inserts a sheet metal jog in the current model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalJog( _  
    ByVal Angle As System.Double, _  
    ByVal Radius As System.Double, _  
    ByVal OffsetDist As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal FixProjLen As System.Boolean, _  
    ByVal DimPos As System.Short, _  
    ByVal BendPos As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Angle As System.Double  
Dim Radius As System.Double  
Dim OffsetDist As System.Double  
Dim FlipDir As System.Boolean  
Dim FixProjLen As System.Boolean  
Dim DimPos As System.Short  
Dim BendPos As System.Short
```

```
instance.InsertSheetMetalJog(Angle, Radius, OffsetDist, FlipDir, FixProjLen, DimPos, BendPos)
```

C#

```
void InsertSheetMetalJog(  
    System.double Angle,  
    System.double Radius,  
    System.double OffsetDist,  
    System.bool FlipDir,  
    System.bool FixProjLen,  
    System.short DimPos,  
    System.short BendPos  
)
```

C++/CLI

```

void InsertSheetMetalJog(
& System.double Angle,
& System.double Radius,
& System.double OffsetDist,
& System.bool FlipDir,
& System.bool FixProjLen,
& System.short DimPos,
& System.short BendPos
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Jog angle

Radius

Jog radius

OffsetDist

Offset distance

FlipDir

True flips the jog direction, false does not

FixProjLen

True fixes the projected length, false does not

DimPos

Dimension position

BendPos

Bend position

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalJog](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IJogFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertSheetMetalMiterFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalMiterFlange Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalMiterFlange](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertSheetMetalMiterFlange( _
    ByVal UseReliefRatio As System.Boolean, _
    ByVal UseDefaultGap As System.Boolean, _
    ByVal UseAutoRelief As System.Boolean, _
    ByVal GlobalRadius As System.Double, _
    ByVal RipGap As System.Double, _
    ByVal AutoReliefRatio As System.Double, _
    ByVal AutoReliefWidth As System.Double, _
    ByVal AutoReliefDepth As System.Double, _
    ByVal ReliefType As System.Integer, _
    ByVal RipLocation As System.Integer, _
    ByVal TrimSideBends As System.Boolean _
)

```

```

Dim instance As IModelDoc2
Dim UseReliefRatio As System.Boolean
Dim UseDefaultGap As System.Boolean
Dim UseAutoRelief As System.Boolean
Dim GlobalRadius As System.Double
Dim RipGap As System.Double
Dim AutoReliefRatio As System.Double
Dim AutoReliefWidth As System.Double
Dim AutoReliefDepth As System.Double
Dim ReliefType As System.Integer
Dim RipLocation As System.Integer
Dim TrimSideBends As System.Boolean

```

```

instance.InsertSheetMetalMiterFlange(UseReliefRatio, UseDefaultGap, UseAutoRelief, GlobalRadius,
RipGap, AutoReliefRatio, AutoReliefWidth, AutoReliefDepth, ReliefType, RipLocation, TrimSideBends)

```

C#

```

void InsertSheetMetalMiterFlange(
    System.bool UseReliefRatio,
    System.bool UseDefaultGap,
    System.bool UseAutoRelief,
    System.double GlobalRadius,
    System.double RipGap,
    System.double AutoReliefRatio,
    System.double AutoReliefWidth,
    System.double AutoReliefDepth,
    System.int ReliefType,
    System.int RipLocation,
    System.bool TrimSideBends
)

```

C++/CLI

```

void InsertSheetMetalMiterFlange(
    & System.bool UseReliefRatio,
    & System.bool UseDefaultGap,
    & System.bool UseAutoRelief,
    & System.double GlobalRadius,
    & System.double RipGap,
    & System.double AutoReliefRatio,
    & System.double AutoReliefWidth,
    & System.double AutoReliefDepth,
    & System.int ReliefType,
    & System.int RipLocation,
    & System.bool TrimSideBends
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseReliefRatio

UseDefaultGap

UseAutoRelief

GlobalRadius

RipGap

AutoReliefRatio

AutoReliefWidth

AutoReliefDepth

ReliefType

RipLocation

TrimSideBends

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalMiterFlange](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalUnfold Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalUnfold Method (IModelDoc2)

Inserts an unfold feature at the selected objects.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalUnfold()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertSheetMetalUnfold()
```

C#

```
void InsertSheetMetalUnfold()
```

C++/CLI

```
void InsertSheetMetalUnfold();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalUnfold](#).

Example

[Insert and Access Fold Feature \(C#\)](#)
[Insert and Access Fold Feature \(VB.NET\)](#)
[Insert and Access Fold Feature \(VBA\)](#)

Remarks

Select the objects for the unfold feature before calling this method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertSketch Method (IModelDoc2)

[See Also](#)



Obsolete. Superseded by [ISketchManager::InsertSketch](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSketch()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertSketch()
```

C#

```
void InsertSketch()
```

C++/CLI

```
void InsertSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketch](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSketch2 Method (IModelDoc2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IModelDoc2 Interface](#) : InsertSketch2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::InsertSketch](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSketch2( _  
    ByVal UpdateEditRebuild As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UpdateEditRebuild As System.Boolean  
instance.InsertSketch2(UpdateEditRebuild)
```

C#

```
void InsertSketch2(  
    System.bool UpdateEditRebuild  
)
```

C++/CLI

```
void InsertSketch2(  
    & System.bool UpdateEditRebuild  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpdateEditRebuild

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketch2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchForEdgeFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchForEdgeFlange Method (IModelDoc2)

Inserts a profile sketch of an edge flange in this sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchForEdgeFlange( _  
    ByVal FlangeEdge As System.Object, _  
    ByVal DAngle As System.Double, _  
    ByVal FlipDir As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim FlangeEdge As System.Object  
Dim DAngle As System.Double  
Dim FlipDir As System.Boolean  
Dim value As System.Object
```

```
value = instance.InsertSketchForEdgeFlange(FlangeEdge, DAngle, FlipDir)
```

C#

```
System.object InsertSketchForEdgeFlange(  
    System.object FlangeEdge,  
    System.double DAngle,  
    System.bool FlipDir  
)
```

C++/CLI

```
System.Object^ InsertSketchForEdgeFlange(  
& System.Object^ FlangeEdge,  
& System.double DAngle,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdge

Edge with which to create an edge flange

DAngle

Angle of flange

FlipDir

True reverses the offset direction of the flange, false does not

Return Value

Sketch for the edge flange, returned as a [feature](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchForEdgeFlange](#).

Example

See the [IEdgeFlangeFeatureData](#) examples.

Remarks

After calling this method, you must create the profile for the flange on the appropriate plane. Then use [IFeatureManager::CreateDefinition](#), [IEdgeFlangeFeatureData](#), and [IFeatureManager::CreateFeature](#) to create the edge flange.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertSketchPicture Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchPicture Method (IModelDoc2)

Inserts a picture into the current sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchPicture( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.InsertSketchPicture(FileName)
```

C#

```
System.bool InsertSketchPicture(  
    System.string FileName  
)
```

C++/CLI

```
System.bool InsertSketchPicture(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName
Path to image file including file extension

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchPicture](#).

Remarks

Supported image types are:

- Windows bitmap (*.bmp)
- Tagged Image Format (*.tif)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertSketchPictureData Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchPictureData Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchPictureData Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertSketchPictureData64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Inserts a picture into the current sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSketchPictureData( _  
    ByVal Width As System.Short, _  
    ByVal Height As System.Short, _  
    ByVal PDataIn As System.Integer _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Width As System.Short  
Dim Height As System.Short  
Dim PDataIn As System.Integer
```

```
instance.InsertSketchPictureData(Width, Height, PDataIn)
```

C#

```
void InsertSketchPictureData(  
    System.short Width,  
    System.short Height,  
    System.int PDataIn  
)
```

C++/CLI

```
void InsertSketchPictureData(  
& System.short Width,  
& System.short Height,  
& System.int PDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Width

Width of the image

Height

Height of the image

PDataIn

Pointer to a raw bitmap (HBITMAP) cast as a long

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchPictureData](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

A copy of the specified bitmap data is created. The SOLIDWORKS Design software does not release your bitmap object.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::InsertSketchPicture Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchPictureDatax64 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchPictureDatax64 Method (IModelDoc2)

Inserts a picture into the current sketch in 64-bit applications.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSketchPictureDatax64( _  
    ByVal Width As System.Integer, _  
    ByVal Height As System.Integer, _  
    ByVal PDataIn As System.Long _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Width As System.Integer  
Dim Height As System.Integer  
Dim PDataIn As System.Long
```

```
instance.InsertSketchPictureDatax64(Width, Height, PDataIn)
```

C#

```
void InsertSketchPictureDatax64(  
    System.int Width,  
    System.int Height,  
    System.long PDataIn  
)
```

C++/CLI

```
void InsertSketchPictureData64(
& System.int Width,
& System.int Height,
& System.int64 PDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Width

Width of the image

Height

Height of the image

PDataIn

Pointer to a raw bitmap (HBITMAP) cast as a LONGLONG

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchPictureData64](#).

Remarks

This method is only available through early binding and with 64-bit versions of the SOLIDWORKS Design software.

A copy of the specified bitmap data is created. The SOLIDWORKS Design software does not release your bitmap object.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::InsertSketchPictureData Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSketchText Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchText Method (IModelDoc2)

Inserts sketch text.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchText(
    ByVal Ptx As System.Double, _
    ByVal Pty As System.Double, _
    ByVal Ptz As System.Double, _
    ByVal Text As System.String, _
    ByVal Alignment As System.Integer, _
    ByVal FlipDirection As System.Integer, _
    ByVal HorizontalMirror As System.Integer, _
    ByVal WidthFactor As System.Integer, _
    ByVal SpaceBetweenChars As System.Integer _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim Ptx As System.Double
Dim Pty As System.Double
Dim Ptz As System.Double
Dim Text As System.String
```



```

Dim Alignment As System.Integer
Dim FlipDirection As System.Integer
Dim HorizontalMirror As System.Integer
Dim WidthFactor As System.Integer
Dim SpaceBetweenChars As System.Integer
Dim value As System.Object

```

```

value = instance.InsertSketchText(Ptx, Pty, Ptz, Text, Alignment, FlipDirection, HorizontalMirror,
WidthFactor, SpaceBetweenChars)

```

C#

```

System.object InsertSketchText(
    System.double Ptx,
    System.double Pty,
    System.double Ptz,
    System.string Text,
    System.int Alignment,
    System.int FlipDirection,
    System.int HorizontalMirror,
    System.int WidthFactor,
    System.int SpaceBetweenChars
)

```

C++/CLI

```

System.Object^ InsertSketchText(
    & System.double Ptx,
    & System.double Pty,
    & System.double Ptz,
    & System.String^ Text,
    & System.int Alignment,
    & System.int FlipDirection,
    & System.int HorizontalMirror,
    & System.int WidthFactor,
    & System.int SpaceBetweenChars
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Ptx

X coordinate of starting point of text block

Pty

Y coordinate of starting point of text block

Ptz

Z coordinate of starting point of text block

Text

Text to insert (see **Remarks**)

Alignment

- 0 = Left
- 1 = Center
- 2 = Right
- 3 = Fully justified

(see **Remarks**)

FlipDirection

1 to flip text vertically about the selected entity, 0 to not (see **Remarks**)

HorizontalMirror

1 to flip text horizontally, 0 to not

WidthFactor

6 <= Percentage by which to evenly widen each character in the text block <= 1667

SpaceBetweenChars

1 <= Percentage of space between each character in the text block <= 10000; valid only if Alignment != 3

Return Value

[Sketch text](#)

Example

[Insert Text At Angle \(VBA\)](#)
[Align Text With Line \(VBA\)](#)
[Insert Sketch Text and Hole \(VBA\)](#)
[Insert Sketch Text and Hole \(VB.NET\)](#)
[Insert Sketch Text and Hole \(C#\)](#)

Remarks

Alignment and FlipDirection are valid only when a curve, edge, or sketch segment is selected. Text appears along the selected entity. If an entity is not selected, Text appears horizontally starting at the origin. [Select](#) the curve, edge, or sketch segment with Mark = 1.

See the SOLIDWORKS Design user-interface **help > Sketching > Sketch Entities > Sketch Text > SketchText PropertyManager** topic for more information about this functionality.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DissolveSketchText Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSplinePoint Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSplinePoint Method (IModelDoc2)

Inserts a spline point.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSplinePoint( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
  
instance.InsertSplinePoint(X, Y, Z)
```

C#

```
void InsertSplinePoint(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
void InsertSplinePoint(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of spline point

Y
Y coordinate of spline point

Z
Z coordinate of spline point

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSplinePoint](#).

Example

[Insert Spline Point \(VBA\)](#)
[Insert Spline Point \(VB.NET\)](#)
[Insert Spline Point \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateSpline Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSplitLineProject Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSplitLineProject Method (IModelDoc2)

Splits a face by projecting sketch lines onto the face.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSplitLineProject( _  
    ByVal IsDirectional As System.Boolean, _  
    ByVal FlipDir As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim IsDirectional As System.Boolean  
Dim FlipDir As System.Boolean  
  
instance.InsertSplitLineProject(IsDirectional, FlipDir)
```

C#

```
void InsertSplitLineProject(  
    System.bool IsDirectional,  
    System.bool FlipDir  
)
```

C++/CLI

```
void InsertSplitLineProject(  
& System.bool IsDirectional,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IsDirectional

Whether to project in one direction:

- 0 projects in both directions
- 1 projects in one direction

FlipDir

Whether to project along the normal to the sketch plane; valid only when isDirectional = 1:

- 0 projects in a direction opposite to the normal of the sketch plane
- 1 projects along the normal of the sketch plane

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSplitLineProject](#).

Example

[Create Projection Split Line Feature \(VBA\)](#)
[Create Projection Split Line Feature \(VB.NET\)](#)
[Create Projection Split Line Feature \(C#\)](#)

Remarks

- The sketch to project must be selected using [IModelDocExtension::SelectByID2](#) with a mark value of 4.
- The faces to split must be selected using [IModelDocExtension::SelectByID2](#) with mark values of 1.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertSplitLineSil Method](#)
[ISplitLineFeatureData Interface](#)
[IFeatureManager::InsertSplitLineIntersect Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSplitLineSil Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSplitLineSil Method (IModelDoc2)

Splits a face by creating split lines along the silhouette of the selected faces.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSplitLineSil()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertSplitLineSil()
```

C#

```
void InsertSplitLineSil()
```

C++/CLI

```
void InsertSplitLineSil();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSplitLineSil](#).

Remarks

The silhouette curves differ based on your orientation. Therefore, this method requires a selection to specify the desired direction.

Valid items to select to specify the direction are:

- reference plane
- planar face
- edge
- axis
- two points

The item selected for direction must be selected using [IModelDocExtension::SelectByID2](#) with a mark value of 2. The faces to split must be selected using [IModelDocExtension::SelectByID2](#) with mark values of 1.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertSplitLineProject Method](#)
[ISplitLineFeatureData Interface](#)
[IFeatureManager::InsertSplitLineIntersect Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertStackedBalloon Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertStackedBalloon Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertStackedBalloon](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStackedBalloon( _  
    ByVal Style As System.Integer, _  
    ByVal Size As System.Integer, _  
    ByVal UpperTextStyle As System.Integer, _  
    ByVal UpperText As System.String, _  
    ByVal LowerTextStyle As System.Integer, _  
    ByVal LowerText As System.String _  
) As Note
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Style As System.Integer  
Dim Size As System.Integer  
Dim UpperTextStyle As System.Integer  
Dim UpperText As System.String  
Dim LowerTextStyle As System.Integer  
Dim LowerText As System.String  
Dim value As Note
```

```
value = instance.InsertStackedBalloon(Style, Size, UpperTextStyle, UpperText, LowerTextStyle,  
LowerText)
```

C#

```
Note InsertStackedBalloon(
    System.int Style,
    System.int Size,
    System.int UpperTextStyle,
    System.string UpperText,
    System.int LowerTextStyle,
    System.string LowerText
)
```

C++/CLI

```
Note^ InsertStackedBalloon(
    & System.int Style,
    & System.int Size,
    & System.int UpperTextStyle,
    & System.String^ UpperText,
    & System.int LowerTextStyle,
    & System.String^ LowerText
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Style

Balloon style as defined in [swBalloonStyle_e](#)

Size

Balloon size as defined in [swBalloonFit_e](#)

UpperTextStyle

Text style for the upper text of the balloon as defined in [swBalloonTextContent_e](#)

UpperText

Text in the balloon

LowerTextStyle

Text style for the lower text of the balloon as defined in [swBalloonTextContent_e](#)

LowerText

Text in the lower part of the balloon when Style = swBS_SplitCirc

Return Value

Newly created [note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertStackedBalloon](#).

Remarks

This method adds a single balloon note attached to the pre-selected entity. The note is located near the pre-selection.

This method returns a Note object, which you can then use to access the note; for example, setting the font of the note text or setting the position of the Note. You can stack notes on the Note object returned by this method.

Use [INote::GetBalloonStack](#) with this note to get the [IBalloonStack](#) object. You can control various properties of the stack, such as the direction and size. You can also use the BalloonStack to stack more balloons on this first balloon.

If the balloon style is split circle, both the lower- and upper-text arguments are used. If the balloon style is anything other than split circle, the upper-text arguments are used and the lower-text arguments are ignored.

If the text style is item number or quantity, the note text is determined through the pre-selected entity to which this note is attached and the corresponding text argument is ignored. If the pre-selection is just a location on the drawing, not an entity, then the text style should be custom and you must specify the text to use.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[INote::IsStackedBalloon Method](#)

[INote::IsStackedBalloonMaster Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

InsertSurfaceFinishSymbol2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSurfaceFinishSymbol2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertSurfaceFinishSymbol3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSurfaceFinishSymbol2( _  
    ByVal SymType As System.Integer, _  
    ByVal LeaderType As System.Integer, _  
    ByVal LocX As System.Double, _  
    ByVal LocY As System.Double, _  
    ByVal LocZ As System.Double, _  
    ByVal LaySymbol As System.Integer, _  
    ByVal ArrowType As System.Integer, _  
    ByVal MachAllowance As System.String, _  
    ByVal OtherVals As System.String, _  
    ByVal ProdMethod As System.String, _  
    ByVal SampleLen As System.String, _  
    ByVal MaxRoughness As System.String, _  
    ByVal MinRoughness As System.String, _  
    ByVal RoughnessSpacing As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim SymType As System.Integer  
Dim LeaderType As System.Integer  
Dim LocX As System.Double  
Dim LocY As System.Double  
Dim LocZ As System.Double  
Dim LaySymbol As System.Integer  
Dim ArrowType As System.Integer  
Dim MachAllowance As System.String  
Dim OtherVals As System.String  
Dim ProdMethod As System.String  
Dim SampleLen As System.String  
Dim MaxRoughness As System.String  
Dim MinRoughness As System.String  
Dim RoughnessSpacing As System.String  
Dim value As System.Boolean
```

```
value = instance.InsertSurfaceFinishSymbol2(SymType, LeaderType, LocX, LocY, LocZ, LaySymbol,  
ArrowType, MachAllowance, OtherVals, ProdMethod, SampleLen, MaxRoughness, MinRoughness,  
RoughnessSpacing)
```

C#

```
System.bool InsertSurfaceFinishSymbol2(  
    System.int SymType,  
    System.int LeaderType,  
    System.double LocX,  
    System.double LocY,  
    System.double LocZ,  
    System.int LaySymbol,  
    System.int ArrowType,  
    System.string MachAllowance,  
    System.string OtherVals,  
    System.string ProdMethod,  
    System.string SampleLen,  
    System.string MaxRoughness,  
    System.string MinRoughness,  
    System.string RoughnessSpacing  
)
```

```

System.bool InsertSurfaceFinishSymbol2(
& System.int SymType,
& System.int LeaderType,
& System.double LocX,
& System.double LocY,
& System.double LocZ,
& System.int LaySymbol,
& System.int ArrowType,
& System.String^ MachAllowance,
& System.String^ OtherVals,
& System.String^ ProdMethod,
& System.String^ SampleLen,
& System.String^ MaxRoughness,
& System.String^ MinRoughness,
& System.String^ RoughnessSpacing
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymType

LeaderType

LocX

LocY

LocZ

LaySymbol

ArrowType

MachAllowance

OtherVals

ProdMethod

SampleLen

MaxRoughness

MinRoughness

RoughnessSpacing

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSurfaceFinishSymbol2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSweepRefSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSweepRefSurface Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionSweep3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertSweepRefSurface( _
    ByVal Propagate As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean _
)

```

```

Dim instance As IModelDoc2
Dim Propagate As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean

instance.InsertSweepRefSurface(Propagate, TwistCtrlOption, KeepTangency, ForceNonRational)

```

C#

```

void InsertSweepRefSurface(
    System.bool Propagate,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational
)

```

C++/CLI

```

void InsertSweepRefSurface(
    & System.bool Propagate,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

TwistCtrlOption

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSweepRefSurface](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSweepRefSurface2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSweepRefSurface2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager.InsertProtrusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertSweepRefSurface2( _
    ByVal Propagate As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short _
)

```

Visual Basic (Usage)

```

Dim instance As IModelDoc2
Dim Propagate As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short

```

```

instance.InsertSweepRefSurface2(Propagate, TwistCtrlOption, KeepTangency, ForceNonRational,
StartMatchingType, EndMatchingType)

```

C#

```

void InsertSweepRefSurface2(
    System.bool Propagate,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType
)

```

C++/CLI

```

void InsertSweepRefSurface2(
    & System.bool Propagate,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.short StartMatchingType,
    & System.short EndMatchingType
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSweepRefSurface2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertWeldSymbol2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertWeldSymbol2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertWeldSymbol3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertWeldSymbol2( _
    ByVal Dim1 As System.String, _

```

```

ByVal Symbol As System.String, _
ByVal Dim2 As System.String, _
ByVal Symmetric As System.Boolean, _
ByVal FieldWeld As System.Boolean, _
ByVal ShowOtherSide As System.Boolean, _
ByVal DashOnTop As System.Boolean, _
ByVal Peripheral As System.Boolean, _
ByVal HasProcess As System.Boolean, _
ByVal ProcessValue As System.String _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Dim1 As System.String
Dim Symbol As System.String
Dim Dim2 As System.String
Dim Symmetric As System.Boolean
Dim FieldWeld As System.Boolean
Dim ShowOtherSide As System.Boolean
Dim DashOnTop As System.Boolean
Dim Peripheral As System.Boolean
Dim HasProcess As System.Boolean
Dim ProcessValue As System.String

```

```

instance.InsertWeldSymbol2(Dim1, Symbol, Dim2, Symmetric, FieldWeld, ShowOtherSide, DashOnTop,
Peripheral, HasProcess, ProcessValue)

```

C#

```

void InsertWeldSymbol2(
    System.string Dim1,
    System.string Symbol,
    System.string Dim2,
    System.bool Symmetric,
    System.bool FieldWeld,
    System.bool ShowOtherSide,
    System.bool DashOnTop,
    System.bool Peripheral,
    System.bool HasProcess,
    System.string ProcessValue
)

```

C++/CLI

```

void InsertWeldSymbol2(
    & System.String^ Dim1,
    & System.String^ Symbol,
    & System.String^ Dim2,
    & System.bool Symmetric,
    & System.bool FieldWeld,
    & System.bool ShowOtherSide,
    & System.bool DashOnTop,
    & System.bool Peripheral,
    & System.bool HasProcess,
    & System.String^ ProcessValue
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Dim1

Symbol
See **Remarks**.
Dim2

Symmetric

FieldWeld

ShowOtherSide

DashOnTop

Peripheral
HasProcess
ProcessValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertWeldSymbol2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWeldSymbol3 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertWeldSymbol3 Method (IModelDoc2)

Inserts a weld symbol into the model.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldSymbol3() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.InsertWeldSymbol3()
```

C#

```
System.Object InsertWeldSymbol3()
```

C++/CLI

```
System.Object^ InsertWeldSymbol3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [weld symbol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertWeldSymbol3](#).

Example

[Insert Weld Symbol Example \(VBA\)](#)
[Insert Weld Symbol Example \(VB.NET\)](#)
[Insert Weld Symbol Example \(C#\)](#)

Remarks

To use this method, insert the weld symbol into the model and then manipulate the properties and methods on the [IWeldSymbol](#) object.

A list of weld symbol names can be found in **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\lgto1.sym**. The currently supported list of ISO weld symbols is:

- BUTT
- BUSQ
- BUSV

- BUSB
- BUSVBR
- BUSBR
- BUSU
- BUSJ
- BACK
- FILL
- PLUG
- SPOT
- SEAM
- SEAMC
- JSPT
- JSM

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InspectCurvature Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InspectCurvature Method (IModelDoc2)

Adds curvature combs to the selected sketch segment.

.NET Syntax

Visual Basic (Declaration)

```
Sub InspectCurvature()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
instance.InspectCurvature()
```

C#

```
void InspectCurvature()
```

C++/CLI

```
void InspectCurvature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InspectCurvature](#).

Example

[Add or Remove Curvature Combs \(VBA\)](#)

Remarks

See [IModelDoc2::RemoveInspectCurvature](#) to remove curvature combs.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketchSpline::ShowCurvatureCombs Property](#)
[IModelDoc2::RemoveInspectCurvature Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IPageSetup Property (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IPageSetup Property (IModelDoc2)

Gets the page setup for this document.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IPageSetup As PageSetup

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As PageSetup
```

```
value = instance.IPageSetup
```

C#

```
PageSetup IPageSetup {get;}
```

C++/CLI

```
property PageSetup^ IPageSetup {  
    PageSetup^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Page setup](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IPageSetup](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::PageSetup Property](#)
[IModelDocExtension::AppPageSetup Property](#)
[IModelDocExtension::UsePageSetup Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IParameter Method (IModelDoc2)

[See Also](#)



Gets the specified parameter.

.NET Syntax

Visual Basic (Declaration)

```
Function IParameter( _
    ByVal StringIn As System.String _
) As Dimension
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim StringIn As System.String
Dim value As Dimension

value = instance.IParameter(StringIn)
```

C#

```
Dimension IParameter(
    System.string StringIn
)
```

C++/CLI

```
Dimension^ IParameter(
    & System.String^ StringIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StringIn

Name of parameter (for example, "D1@Base-Revolve")

Return Value

[Dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IParameter](#).

Remarks

Most parameters are [dimensions](#).

The StringIn argument must be the fully qualified dimension name (for example, "D1@Base-Extrude") for this method. However, the full dimension name is not needed if you use [IFeature::IParameter](#) or [IFeature::Parameter](#).

Some characters are recognized as special characters. The use of these characters in names or parts or features can cause this method to fail to return a dimension. These special characters are not recognized in names of parts or features:

- at sign (@)
- period (.)
- backslash (/)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::Parameter Method](#)
[IModelDoc2::SetParamValue Method](#)

Availability

IRayIntersections Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IRayIntersections Method (IModelDoc2)**Obsolete.** Superseded by [IModelDocExtension::RayIntersections](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IRayIntersections( _
    ByRef BodiesIn As Body2, _
    ByVal NumBodies As System.Integer, _
    ByRef BasePointsIn As System.Double, _
    ByRef VectorsIn As System.Double, _
    ByVal NumRays As System.Integer, _
    ByVal Options As System.Integer, _
    ByVal HitRadius As System.Double, _
    ByVal Offset As System.Double _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim BodiesIn As Body2
Dim NumBodies As System.Integer
Dim BasePointsIn As System.Double
Dim VectorsIn As System.Double
Dim NumRays As System.Integer
Dim Options As System.Integer
Dim HitRadius As System.Double
Dim Offset As System.Double
Dim value As System.Integer

value = instance.IRayIntersections(BodiesIn, NumBodies, BasePointsIn, VectorsIn, NumRays, Options, HitRadius, Offset)
```

C#

```
System.int IRayIntersections(
    ref Body2 BodiesIn,
    System.int NumBodies,
    ref System.double BasePointsIn,
    ref System.double VectorsIn,
    System.int NumRays,
    System.int Options,
    System.double HitRadius,
    System.double Offset
)
```

C++/CLI

```
System.int IRayIntersections(
    & Body2^% BodiesIn,
    & System.int NumBodies,
    & System.double% BasePointsIn,
    & System.double% VectorsIn,
    & System.int NumRays,
    & System.int Options,
    & System.double HitRadius,
    & System.double Offset
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*BodiesIn*Array [bodies](#) that are hit by the rays (see **Remarks**)

NumBodies

Number of bodies in the BodiesIn array

BasePointsIn

Array containing the x,y,z base points of each ray

VectorsIn

Array containing the direction vectors of each ray

NumRays

Number of rays specified; this should be equal to the number of elements in the (basePointsIn / 3) or (vectorsIn / 3) arrays

Options

Number of options as defined in [swRayPtsOpts_e](#)

HitRadius

Radius of hit cylinder used mainly in grazing cases to establish a hit

Offset

Length tolerance to use to establish whether a hit on a face represents the entry or exit of the ray from the body (see **Remarks**)

Return Value

Number of intersections found

Visual Basic for Applications (VBA) Syntax

See [IModelDoc2::IRayIntersections](#).

Remarks

The method performs the intersection operations between the specified bodies and the ray vectors. To get the results (a set of intersection point: intersection normals, and the bodies that were hit from your bodiesIn array) of the intersection operations, you must call [IModelDoc2::GetRayIntersectionsPoints](#) or [IModelDoc2::IGetRayIntersectionPoints](#) and [IModelDoc2::GetRayIntersectionTopology](#) or [IModelDoc2::IGetRayIntersectionsTopology](#).

Information returned by IModelDoc2::GetRayIntersectionsPoints, IModelDoc2::IGetRayIntersectionPoints, IModelDoc2::GetRayIntersectionTopology, and IModelDoc2::IGetRayIntersectionsTopology depends partially on the value of the options argument. Valid values can be concatenated together using bitwise operations. See IModelDoc2::GetRayIntersectionsPoints or IModelDoc2::IGetRayIntersectionPoints to see which data is always output regardless of the values specified in the options argument.

For the COM interface, the return value, the intersection count, must be used in determining the size of arrays to allocate for return values from IModelDoc2::IGetRayIntersectionsPoints and IModelDoc2::IGetRayIntersectionsTopology.

For each ray that hits an edge or a vertex, the offset distance is added in both the positive and negative directions along the ray and the points computed are tested for spatial inclusion in the hit body. This determines if the ray was entering, leaving, or just grazing the body at the hit point. Entry and exit onto faces can be computed does not require such an offset.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IMultiSelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)
[IModelDoc2::SelectByRay Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRelease3rdPartyStorage Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IRelease3rdPartyStorage Method (IModelDoc2)

Releases the specified third-party stream.

.NET Syntax

Visual Basic (Declaration)

```
Sub IRelease3rdPartyStorage( _  
    ByVal StringIn As System.String _  
)
```

```
Dim instance As IModelDoc2
Dim StringIn As System.String

instance.IRelease3rdPartyStorage(StringIn)
```

C#

```
void IRelease3rdPartyStorage(
    System.string StringIn
)
```

C++/CLI

```
void IRelease3rdPartyStorage(
    & System.String^ StringIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StringIn

Name of the stream to release

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IRelease3rdPartyStorage](#).

Remarks

You can use this method with [IModelDoc2::IGet3rdPartyStorage](#). You must use it if you are not using the LoadFromStorageNotify event. You must call this method when the call to IModelDoc2::IGet3rdPartyStorage returns a NULL stream.

NOTE: If you are using serialization, then be careful with the standard MFC macros; otherwise, you can get messages like **Unexpected File Format** after your application is unloaded. One way of using IMPLEMENT_SERIAL is:

```
IMPLEMENT_SERIAL( CCustomAttr, CObject, VERSIONABLE_SCHEMA|0 )
```

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[DAssemblyDocEvents_LoadFromStorageNotifyEventHandler Delegate](#)
[DAssemblyDocEvents_SaveToStorageNotifyEventHandler Delegate](#)
[DDrawingDocEvents_LoadFromStorageNotifyEventHandler Delegate](#)
[DDrawingDocEvents_SaveToStorageNotifyEventHandler Delegate](#)
[DPartDocEvents_LoadFromStorageNotifyEventHandler Delegate](#)
[DPartDocEvents_SaveToStorageNotifyEventHandler Delegate](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsActive Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsActive Method (IModelDoc2)

Gets whether the specified assembly component is shown or hidden in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function IsActive( _
    ByVal CompStr As System.String _
) As System.Boolean
```

```
Dim instance As IModelDoc2
Dim CompStr As System.String
Dim value As System.Boolean

value = instance.IsActive(CompStr)
```

C#

```
System.bool IsActive(
    System.string CompStr
)
```

C++/CLI

```
System.bool IsActive(
    & System.String^ CompStr
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompStr

Name specification of the component

Return Value

True if component is shown, false if component is hidden

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsActive](#).

Remarks

The CompStr parameter is the full assembly component name designation. The format of the name designation is:

parentModel/childModel

where:

childModel is the model whose display status you want to determine. For example, if you want to determine the display status of a part named SPOKE.SLDPRT and if this part is a child of WHEEL.SLDPRT, which itself is a child of AXIS.SLDPRT, then you would specify CompStr as follows:

AXIS/WHEEL/SPOKE

TIP: The assembly component name designation is shown in the lower-left corner of the SOLIDWORKS Design application when an assembly component is selected.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ShowComponent2 Method](#)
[IModelDoc2::HideComponent2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IsEditingSelf Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsEditingSelf Method (IModelDoc2)

Gets whether this model is being edited in the context of another document.

.NET Syntax

Visual Basic (Declaration)

```
Function IsEditingSelf() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Boolean  
  
value = instance.IsEditingSelf()
```

C#

```
System.bool IsEditingSelf()
```

C++/CLI

```
System.bool IsEditingSelf();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the model document is the active edit target, false if the model document is not the active edit target of this SOLIDWORKS Design session

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsEditingSelf](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IAssemblyDoc::EditPart2 Method](#)
[IAssemblyDoc::IGetEditTarget2 Method](#)
[IAssemblyDoc::GetEditTarget Method](#)
[IAssemblyDoc::GetEditTargetComponent Method](#)
[IAssemblyDoc::EditAssembly Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISelectByRay Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ISelectByRay Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SelectByRay](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISelectByRay( _  
    ByRef PointIn As System.Double, _  
    ByRef VectorIn As System.Double, _  
    ByVal RadiusIn As System.Double, _  
    ByVal TypeWanted As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PointIn As System.Double  
Dim VectorIn As System.Double
```

```
Dim RadiusIn As System.Double
Dim TypeWanted As System.Integer
Dim value As System.Boolean

value = instance.ISelectByRay(PointIn, VectorIn, RadiusIn, TypeWanted)
```

C#

```
System.bool ISelectByRay(
    ref System.double PointIn,
    ref System.double VectorIn,
    System.double RadiusIn,
    System.int TypeWanted
)
```

C++/CLI

```
System.bool ISelectByRay(
    & System.double% PointIn,
    & System.double% VectorIn,
    & System.double RadiusIn,
    & System.int TypeWanted
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointIn

Array containing 3 doubles that define the start point of the ray

VectorIn

Array containing 3 doubles that define the direction of the ray

RadiusIn

Radius of the ray in meters

TypeWanted

Type of objects to select as defined in [swSelectType_e](#)

Return Value

True if an object was selected, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISelectByRay](#).

Remarks

This method:

- Defines a cylindrical region of infinite length that begins at pointIn, runs parallel to VectorIn, and has a radius of RadiusIn. If you specify edge or vertex entities, then the first edge or vertex found within the selection cylinder is selected.
- Only selects entity objects, which include faces, edges, vertices, and so on. You cannot select sketch objects using this method.
- Does not support the selection of silhouette edges.

Only a single entity is selected within the distance radius regardless of multiple entities existing within that radius. For selecting face entities, RadiusIn is ignored and a cylinder of infinitely small radius is used.

Use model space view to determine the selection vector in a drawing.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetRayIntersectionsPoints Method](#)
[IModelDoc2::GetRayIntersectionsTopology Method](#)
[IModelDoc2::IGetRayIntersectionsPoints Method](#)
[IModelDoc2::IGetRayIntersectionsTopology Method](#)
[IModelDoc2::IMultiSelectByRay Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)
[IModelDoc2::SelectByRay Method](#)

Availability

ISelectionManager Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ISelectionManager Property (IModelDoc2)

Gets the [ISelectionMgr](#) object for this document, which makes the currently selected object available.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property ISelectionManager As SelectionMgr

Visual Basic (Usage)



Dim instance As [IModelDoc2](#)

Dim value As SelectionMgr

value = instance.ISelectionManager

C#

SelectionMgr ISelectionManager {get;}

C++/CLI

```
property SelectionMgr^ ISelectionManager {
    SelectionMgr^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[ISelectionMgr](#) object

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISelectionManager](#).

Example

[Access Assembly \(C++\)](#)
[Deselect Every Other Selected Object \(C++\)](#)
[Display of Item in FeatureManager Design Tree \(C++\)](#)
[Get Curve Spline Points \(C++\)](#)
[Get Parent Features \(C++\)](#)
[Get Selected Feature \(C++\)](#)
[Get Sketch Segment Constraints \(C++\)](#)
[Get Spline Points \(C++\)](#)

Remarks

[ISelectionMgr](#) objects are transient because they are invalid as soon as another selection is made. So, do not hold on to these pointers for any length of time.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SelectionManager Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ISetAngularUnits Method (IModelDoc2)

[See Also](#)



Sets the current angular units.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub ISetAngularUnits( _  
    ByVal UType As System.Short, _  
    ByVal FractBase As System.Short, _  
    ByVal FractDenom As System.Short, _  
    ByVal SigDigits As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UType As System.Short  
Dim FractBase As System.Short  
Dim FractDenom As System.Short  
Dim SigDigits As System.Short  
  
instance.ISetAngularUnits(UType, FractBase, FractDenom, SigDigits)
```

C#

```
void ISetAngularUnits(  
    System.short UType,  
    System.short FractBase,  
    System.short FractDenom,  
    System.short SigDigits  
)
```

C++/CLI

```
void ISetAngularUnits(  
& System.short UType,  
& System.short FractBase,  
& System.short FractDenom,  
& System.short SigDigits  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UType

Angular units as defined in [swAngleUnit_e](#)

FractBase

Not supported; however, input to this field is required but is not used

FractDenom

Not supported; however, input to this field is required but is not used

SigDigits

Significant digits if using decimal units

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISetAngularUnits](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetAngularUnits Method](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::GetUnits Method](#)
[IModelDoc2::GetUserUnit Method](#)
[IModelDoc2::IGetUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)

[IModelDoc2::SetUnits Method](#)
[IModelDoc2::LengthUnit Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetNextSelectionGroupId Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ISetNextSelectionGroupId Method (IModelDoc2)

Sets the group ID for all remaining selections.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetNextSelectionGroupId( _  
    ByVal ID As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ID As System.Integer  
  
instance.ISetNextSelectionGroupId(ID)
```

C#

```
void ISetNextSelectionGroupId(  
    System.int ID  
)
```

C++/CLI

```
void ISetNextSelectionGroupId(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Group ID

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISetNextSelectionGroupId](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetUserPreferenceTextFormat Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ISetUserPreferenceTextFormat Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SetUserPreferenceTextFormat](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ISetUserPreferenceTextFormat( _  
    ByVal UserPreferenceValue As System.Integer, _  
    ByVal Value As TextFormat _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UserPreferenceValue As System.Integer  
Dim Value As TextFormat  
Dim value As System.Boolean  
  
value = instance.ISetUserPreferenceTextFormat(UserPreferenceValue, Value)
```

C#

```
System.bool ISetUserPreferenceTextFormat(  
    System.int UserPreferenceValue,  
    TextFormat Value  
)
```

C++/CLI

```
System.bool ISetUserPreferenceTextFormat(  
& System.int UserPreferenceValue,  
& TextFormat^ Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPreferenceValue

User preference to change as defined in [swUserPreferenceTextFormat_e](#)

Value

Text format to which to assign the user preference specified in UserPreferenceValue

Return Value

True if the setting is changed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISetUserPreferenceTextFormat](#).

Remarks

This method is

- intended for setting detailing text formats.
- equivalent to interactively setting document options in the SOLIDWORKS Design software.

See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsExploded Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::IsExploded](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IsExploded() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.IsExploded()
```

C#

```
System.bool IsExploded()
```

C++/CLI

```
System.bool IsExploded();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the model is currently exploded, false if the model is collapsed

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsExploded](#).

Remarks

To take advantage of the explode mechanism, see [IAssemblyDoc::ShowExploded2](#), [IAssemblyDoc::CreateExplodedView](#), [IAssemblyDoc::GetExplodedViewNames](#), [IAssemblyDoc::IGetExplodedViewNames](#), [IAssemblyDoc::GetExplodedViewCount](#), [IAssemblyDoc::AutoExplode](#), [IAssemblyDoc::ViewExplodeAssembly](#), [IAssemblyDoc::ViewCollapseAssembly](#), [IView::ShowExploded](#), and [IView::IsExploded](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISketchSplineByEqnParams Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ISketchSplineByEqnParams Method (IModelDoc2)

Creates a spline on the active 2D sketch using the specified b-curve parameters.

.NET Syntax

Visual Basic (Declaration)

```
Function ISketchSplineByEqnParams( _
    ByRef PropArray As System.Integer, _
    ByRef KnotsArray As System.Double, _
```

```
ByRef CntrlPntCoordArray As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PropArray As System.Integer  
Dim KnotsArray As System.Double  
Dim CntrlPntCoordArray As System.Double  
Dim value As System.Integer  
  
value = instance.ISketchSplineByEqnParams(PropArray, KnotsArray, CntrlPntCoordArray)
```

C#

```
System.int ISketchSplineByEqnParams(  
    ref System.int PropArray,  
    ref System.double KnotsArray,  
    ref System.double CntrlPntCoordArray  
)
```

C++/CLI

```
System.int ISketchSplineByEqnParams(  
& System.int% PropArray,  
& System.double% KnotsArray,  
& System.double% CntrlPntCoordArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropArray

Includes dimension, order, number of control points, and periodicity

KnotsArray

knot1, knot2, and so on

CntrlPntCoordArray

controlpoint1[dimension], controlpoint2[dimension], and so on

Return Value

True if created successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ISketchSplineByEqnParams](#).

Example

[Insert Spline in Drawing \(C++\)](#)

Remarks

The PropArray argument contains 4 integers packed into the first two doubles in the array:

- Dimension
- Order
- Number of Control Points
- Periodicity (True or false)

The KnotsArray argument is an array of doubles with (Number of Control Points + Order) elements.

The size of the CntrlPntCoordArray array is based upon the curve dimension:

- Dimension = 2 then each control point is an array of 2 doubles (x, y)
- Dimension = 3 then each control point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each control point is an array of 4 doubles (x, y, z, w) where w = weight

The parameters are provided as 3 arrays, which for COM applications are passed separately.

Pass control point coordinates to this method in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your b-curve parameters to sketch space with the help of [ISketch::ModelToSketchTransform](#).

NOTE: If the spline being generated is a closed spline, then it must be flagged as periodic. In addition, splines generated in sketches must be G continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchSplineByEqnParams2 Method](#)
[IModelDoc2::CreateSplineByEqnParams Method](#)
[IModelDoc2::CreateSplinesByEqnParams Method](#)
[IModelDoc2::ICreateSplineByEqnParams Method](#)
[IModelDoc2::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsLightLockedToModel Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsLightLockedToModel Method (IModelDoc2)

Gets whether the specified light is fixed.

.NET Syntax

Visual Basic (Declaration)

```
Function IsLightLockedToModel( _  
    ByVal LightId As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim LightId As System.Integer  
Dim value As System.Boolean  
  
value = instance.IsLightLockedToModel(LightId)
```

C#

```
System.bool IsLightLockedToModel(  
    System.int LightId  
)
```

C++/CLI

```
System.bool IsLightLockedToModel(  
& System.int LightId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LightId
Light ID

Return Value

True if the light is fixed, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsLightLockedToModel](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::LockLightToModel Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsOpenedReadOnly Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsOpenedReadOnly Method (IModelDoc2)

Gets whether a SOLIDWORKS document is open in read-only mode.

.NET Syntax

Visual Basic (Declaration)

```
Function IsOpenedReadOnly() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.IsOpenedReadOnly()
```

C#

```
System.bool IsOpenedReadOnly()
```

C++/CLI

```
System.bool IsOpenedReadOnly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this document is read-only, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsOpenedReadOnly](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IsOpenedViewOnly Method](#)
[IModelDoc2::SetReadOnlyState Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsOpenedViewOnly Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsOpenedViewOnly Method (IModelDoc2)

Gets whether a SOLIDWORKS document is open in view-only mode.

.NET Syntax

Visual Basic (Declaration)

```
Function IsOpenedViewOnly() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.IsOpenedViewOnly()
```

C#

```
System.bool IsOpenedViewOnly()
```

C++/CLI

```
System.bool IsOpenedViewOnly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if document is in a view-only state, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsOpenedViewOnly](#).

Remarks

Files are loaded using multi-threading. When a file is loading, the graphics are displayed immediately and the end-user is able to perform certain view zooming and panning functions. Until all data and references are loaded, the file is in view-only mode.

A file can be in view-only mode if the end-user chose to load the file for viewing purposes. Your application should check for view-only mode to determine how to proceed.

NOTE: When a file is in view-only mode, many API queries return NULL or empty data.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IsOpenedReadOnly Method](#)
[IModelDoc2::SetReadOnlyState Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsTessellationValid Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IsTessellationValid Method (IModelDoc2)

Gets whether the current set of facets is valid.

.NET Syntax

Visual Basic (Declaration)

```
Function IsTessellationValid() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.IsTessellationValid()
```

C#

```
System.bool IsTessellationValid()
```

C++/CLI

```
System.bool IsTessellationValid();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the facet data is valid; false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IsTessellationValid](#).

Remarks

This method captures operations that invalidate the current set of facets, but does not send a RegenNotify event. For example, if the user changes the rendering tolerance, the RegenNotify event is not sent. However, the current set of facets would be invalid. This action triggers a RepaintNotify from which you can call this method before attempting to use your current set of facet data.

If false is returned, then valid facet information would not be available until SOLIDWORKS completes a repaint operation (RepaintPostNotify). In other words, SOLIDWORKS does not have any valid facet information at this time, and any facet data obtained in earlier calls is invalid.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ITessellation Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IVersionHistory Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IVersionHistory Method (IModelDoc2)

Gets an array of strings indicating the versions in which this model document was saved, including the SOLIDWORKS Design version in which the model document is currently opened and which is the last value returned in the array.

.NET Syntax

Visual Basic (Declaration)

```
Function IVersionHistory() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.String

value = instance.IVersionHistory()
```

C#

```
System.string IVersionHistory()
```

C++/CLI

```
System.String^ IVersionHistory();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of strings showing the history of the document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IVersionHistory](#).

Example

[Document Version History \(C++\)](#)

Remarks

There is one array entry for each major release of the SOLIDWORKS Design software in which the document has been saved. The format for each entry is a major release code followed by one or more minor release codes separated by commas:

<major release code>[<minor release code>]

- or -

<major release code>[<minor release code>,<minor release code>...]

where <major release code> equals a number that remains constant through a major release of the SOLIDWORKS software. For example, the following values are returned based on the corresponding major SOLIDWORKS version:

SOLIDWORKS Release	Version Number
SOLIDWORKS 95	44
SOLIDWORKS 96	243
SOLIDWORKS 97	483
SOLIDWORKS 97Plus	629
SOLIDWORKS 98	822
SOLIDWORKS 98Plus	1008
SOLIDWORKS 99	1137
SOLIDWORKS 2000	1500
SOLIDWORKS 2001	1750
SOLIDWORKS 2001Plus	1950
SOLIDWORKS 2003	2200
SOLIDWORKS 2004	2500
SOLIDWORKS 2005	2800
SOLIDWORKS 2006	3100
SOLIDWORKS 2007	3400
SOLIDWORKS 2008	3800
SOLIDWORKS 2009	4100
SOLIDWORKS 2010	4400
SOLIDWORKS 2011	4700
SOLIDWORKS 2012	5000

SOLIDWORKS 2013	6000
SOLIDWORKS 2014	7000
SOLIDWORKS 2015	8000
SOLIDWORKS 2016	9000
SOLIDWORKS 2017	10000
SOLIDWORKS 2018	11000
SOLIDWORKS 2019	12000
SOLIDWORKS 2020	13000
SOLIDWORKS 2021	14000
SOLIDWORKS 2022	15000
SOLIDWORKS 2023	16000
SOLIDWORKS 2024	17000
SOLIDWORKS 2025	18000
SOLIDWORKS 2026	19000

The <minor release code> equals the year and day of manufacture of a saving version (for example, 1997/320).

Use [IModelDoc2::IGetVersionHistoryCount](#) before calling this method to get the size of the array needed by this method.

To get the version in which a model document was last saved, use [ISldWorks::VersionHistory](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::VersionHistory Method](#)
[ISldWorks::RevisionNumber Method](#)
[IModelDocExtension::IsFutureVersion Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help [Send comments on this topic.](#)

LargeAssemblyMode Property (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : LargeAssemblyMode Property (IModelDoc2)

Gets or sets **Large Assembly Mode** for this document.

.NET Syntax

Visual Basic (Declaration)

Property LargeAssemblyMode As System.Boolean

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim value As System.Boolean

instance.LargeAssemblyMode = value

value = instance.LargeAssemblyMode
```

C#



```
System.bool LargeAssemblyMode {get; set;}
```

C++/CLI

```
property System.bool LargeAssemblyMode {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True enables **Large Assembly Mode**, false disables it

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LargeAssemblyMode](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus SP3, Revision Number 10.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

LBDownAt Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : LBDownAt Method (IModelDoc2)

Generates a left mouse button press (down) event.

.NET Syntax

Visual Basic (Declaration)

```
Sub LBDownAt ( _  
    ByVal Flags As System.Integer, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Flags As System.Integer  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
  
instance.LBDownAt(Flags, X, Y, Z)
```

C#

```
void LBDownAt(  
    System.int Flags,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```

void LBDownAt(
& System.int Flags,
& System.double X,
& System.double Y,
& System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Flags

Any combination of:

- MK_CONTROL
- MK_LBUTTON
- MK_MBUTTON
- MK_RBUTTON
- MK_SHIFT

X

X coordinate of position of cursor

Y

Y coordinate of position of cursor

Z

Z coordinate of position of cursor

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LBDownAt](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::LBUAt Method](#)
[IModelDoc2::DragTo Method](#)
[IMouse Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

LBUpAt Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : LBUpAt Method (IModelDoc2)

Generates a left-mouse button release (up) event.

.NET Syntax

Visual Basic (Declaration)

```

Sub LBUpAt( _
    ByVal Flags As System.Integer, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
)

```

Visual Basic (Usage)

```

Dim instance As IModelDoc2
Dim Flags As System.Integer
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double

```



```
instance.LBUpAt(Flags, X, Y, Z)
```

C#

```
void LBUpAt(  
    System.int Flags,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
void LBUpAt(  
& System.int Flags,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Flags

Any combination of:

- MK_CONTROL
- MK_LBUTTON
- MK_MBUTTON
- MK_RBUTTON
- MK_SHIFT

X

X coordinate of position of cursor

Y

Y coordinate of position of cursor

Z

Z coordinate of position of cursor

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LBUpAt](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::LBDownAt Method](#)
[IModelDoc2::DragTo Method](#)
[IMouse Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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LengthUnit Property (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : LengthUnit Property (IModelDoc2)

Gets and sets the same LengthUnit value used by [IModelDoc2::GetUnits](#), [IModelDoc2::IGetUnits](#), and [IModelDoc2::SetUnits](#).

.NET Syntax

Visual Basic (Declaration)

Property LengthUnit As System.Integer

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim value As System.Integer

instance.LengthUnit = value

value = instance.LengthUnit
```

C#

```
System.int LengthUnit {get; set;}
```

C++/CLI

```
property System.int LengthUnit {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Unit as defined in [swLengthUnit_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LengthUnit](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetAngularUnits Method](#)
[IModelDoc2::IGetAngularUnits Method](#)
[IModelDoc2::IGetUserUnit Method](#)
[IModelDoc2::GetUserUnit Method](#)
[IModelDoc2::ISetAngularUnits Method](#)
[IModelDoc2::SetAngularUnits Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Lights Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Lights Method (IModelDoc2)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub Lights()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2

instance.Lights()
```

C#

```
void Lights()
```

C++/CLI

```
void Lights();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Lights](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

LightSourcePropertyValues Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : LightSourcePropertyValues Property (IModelDoc2)

Gets and sets the light source property values.

.NET Syntax

Visual Basic (Declaration)

```
Property LightSourcePropertyValues( _  
    ByVal ID As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ID As System.Integer  
Dim value As System.Object
```

```
instance.LightSourcePropertyValues(ID) = value
```

```
value = instance.LightSourcePropertyValues(ID)
```

C#

```
System.object LightSourcePropertyValues(  
    System.int ID  
) {get; set;}
```

C++/CLI

```
property System.Object^ LightSourcePropertyValues {  
    System.Object^ get(System.int ID);  
    void set (System.int ID, System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Light source ID

Property Value

Array of 19 doubles containing the light source properties (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LightSourcePropertyValues](#).

Example

Remarks

The light source ID ranges from 0 to n , where n = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

The return values for this property is the following array of 19 doubles:

[*type*, *Intensity*, *Color*[3], *Position*[3], *Direction*[3], *coneAngle*, *falloff*[3], *Ambient*, *Specular*, *isDisabled*, *SpotExponent*]

where:

- *type* = an integer packed into a double that represents the light source type where valid types are taken from openGL definitions (LIGHT_EYE, LIGHT_AMBIENT, LIGHT_SPOT, LIGHT_POINT, LIGHT_DISTANT)
- *Intensity* = Light-source intensity (diffuseness) where values range from 0 to 1.
- *Color*[3] = Light-source color in the form of an array of 3 doubles (R, G, B) where values range from 0 to 1.
- *Position*[3] = Light-source position for spot lights in the form of an array of 3 doubles (X, Y, Z) in model space.
- *Direction*[3] = Light-source direction in the form of an array of 3 doubles (i, j, k)
- *coneAngle* = Light-source cone half angle in degrees where valid values are from 0 to 90 and 180.
- *falloff*[3] = Light-source falloff in the form of an array of 3 doubles (Kc, KI, Kq). These values result in changes in light intensity as the distance between the light position and the vertex change.

NOTE: The falloff[3] values are not supported in SOLIDWORKS 2011 and later.

The result is driven by the following equation:

[$1 / (Kc + D * KI + D * D * Kq)$]

where:

D = the distance between the light's position and the vertex

Kc = constant attenuation

KI = linear attenuation

Kq = quadratic attenuation

- *Ambient* = light-source ambient Intensity
- *Specular* = light-source specular Intensity
- *isDisabled* = an integer packed into a double. True means that the light is disabled and false means the light is not disabled.
- *SpotExponent* = spot exponent

NOTE: The SpotExponent value is not supported in SOLIDWORKS 2011 and later.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDocExtension::GetSunLightSourcePropertyValues Method \(\)](#)
[IModelDocExtension::SetSunLightSourcePropertyValues Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

Gets or sets the light source name that is displayed in the SOLIDWORKS Design user interface.

.NET Syntax

Visual Basic (Declaration)

```
Property LightSourceUserName( _
    ByVal ID As System.Integer _
) As System.String
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim ID As System.Integer
Dim value As System.String

instance.LightSourceUserName(ID) = value

value = instance.LightSourceUserName(ID)
```

C#

```
System.string LightSourceUserName(
    System.int ID
) {get; set;}
```

C++/CLI

```
property System.String^ LightSourceUserName {
    System.String^ get(System.int ID);
    void set (System.int ID, System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Light source ID

Property Value

User name to give to the light source

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::LightSourceUserName](#).

Example

[Turn Lights On \(VBA\)](#)

Remarks

The light source ID ranges from 0 to n , where n = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ListAuxiliaryExternalFileReferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ListAuxiliaryExternalFileReferences Method (IModelDoc2)

Gets the names of auxiliary external file references for this model.

.NET Syntax

Visual Basic (Declaration)

```
Sub ListAuxiliaryExternalFileReferences( _  
    ByRef Feature As System.Object, _  
    ByRef ExternalFileName As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Feature As System.Object  
Dim ExternalFileName As System.Object  
  
instance.ListAuxiliaryExternalFileReferences(Feature, ExternalFileName)
```

C#

```
void ListAuxiliaryExternalFileReferences(  
    out System.object Feature,  
    out System.object ExternalFileName  
)
```

C++/CLI

```
void ListAuxiliaryExternalFileReferences(  
& [Out] System.Object^ Feature,  
& [Out] System.Object^ ExternalFileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Feature

Array containing the names of features that include auxiliary external references

ExternalFileName

Array containing the names of the auxiliary external files

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ListAuxiliaryExternalFileReferences](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IFeatureManager::InsertMacroFeature3](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ListAuxiliaryExternalFileReferencesCount Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ListAuxiliaryExternalFileReferencesCount Method (IModelDoc2)

Gets the number of auxiliary external file references for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function ListAuxiliaryExternalFileReferencesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Integer
```

```
value = instance.ListAuxiliaryExternalFileReferencesCount()
```

C#

```
System.int ListAuxiliaryExternalFileReferencesCount()
```

C++/CLI

```
System.int ListAuxiliaryExternalFileReferencesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of auxiliary file references

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ListAuxiliaryExternalFileReferencesCount](#).

Remarks

Call this method before calling [IModelDoc2::ListExternalFileReferences](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ListExternalFileReferences Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ListExternalFileReferences Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::ListExternalReferences](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ListExternalFileReferences( _  
    ByVal UseSearchRules As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim UseSearchRules As System.Boolean  
Dim value As System.Object  
  
value = instance.ListExternalFileReferences(UseSearchRules)
```

C#

```
System.object ListExternalFileReferences(  
    System.bool UseSearchRules  
)
```

C++/CLI

```
System.Object^ ListExternalFileReferences(  
& System.bool UseSearchRules  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseSearchRules

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ListExternalFileReferences](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)